RICHARD DEAN

SOFTWARE ENGINEER | FRONTEND DEVELOPMENT | LINUX | CYBERSECURITY

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PROFESSIONAL SUMMARY

Hands-on Software Engineer and technology professional combining modern web development with Linux administration, systems troubleshooting, networking, and security-minded engineering.

Experienced building full-stack web applications using React, Next.js, TypeScript, JavaScript, Node.js, Express, PostgreSQL, REST APIs, and Tailwind CSS. Comfortable taking a project beyond the frontend by configuring Linux servers, SSH access, firewalls, reverse proxies, authentication, databases, backups, and self-hosted deployments.

Built and deployed independent technology projects from the application layer through the underlying infrastructure, including 1337 Wing, a self-hosted full-stack e-commerce-style platform incorporating React/Vite, Node.js, Express, PostgreSQL, JWT authentication, bcrypt password hashing, rate limiting, parameterized queries, SSH key authentication, UFW firewall rules, Caddy reverse proxy, Linux, systemd, Tailscale, backups, and security hardening.

Four years of first-line technical support experience at the University of Houston supporting Linux and Windows systems, troubleshooting hardware and software problems, and communicating technical solutions to non-technical users.

Approach to engineering: functional first, security throughout, and design in service of the user.

CORE TECHNICAL SKILLS

Software Development: JavaScript (ES6+) · TypeScript · React · Next.js · Node.js · Express · Python · HTML5 · CSS3 · Tailwind CSS · Redux · REST APIs · Responsive Web Design · Semantic HTML · Full-Stack Development

Backend & Data: PostgreSQL · Firebase · JWT Authentication · bcrypt · Parameterized SQL Queries · Rate Limiting · API Integration

Linux, Systems & Infrastructure: Linux · Ubuntu Server · Bash · SSH · SSH Key Authentication · Caddy · Reverse Proxy Configuration · UFW Firewall · systemd · Server Deployment · Self-Hosting · Raspberry Pi · Tailscale · Backup and Recovery · Linux System Administration

Networking & Security: Cybersecurity · TCP/IP · DNS · Subnetting · Network Troubleshooting · Authentication · Firewall Configuration · Security Hardening · Secure Remote Administration · Security-Minded Web Development

Development Tools: Git · GitHub · Vite · Vercel · npm · VS Code · Linux Command Line

Technical Support: Hardware Troubleshooting · Software Troubleshooting · Operating System Troubleshooting · Windows · Linux · Technical Documentation · User Support · Problem Diagnosis

PROFESSIONAL EXPERIENCE

Technology Support Specialist — *University of Houston — Houston, TX* Sep 2022 – Present

- Provide first-line technical support to students and faculty across Linux and Windows environments.
- Diagnose hardware, software, operating-system, and configuration problems and determine appropriate solutions.
- Troubleshoot technical problems systematically rather than relying solely on predefined fixes.
- Explain technical problems and solutions clearly to users with varying levels of technical knowledge.
- Develop practical troubleshooting skills across real-world computer systems and user environments.

- Support users while balancing technical accuracy, communication, and customer service.
- Apply knowledge of operating systems, networking, hardware, and software to resolve technical issues.

Frontend Developer — *Skinstric AI — Remote* Jul 2026

- Built a real-time AI-powered skin-analysis web application using Next.js and TypeScript.
- Integrated an AI vision API to process user information and return condition results.
- Developed responsive and animated interfaces using Tailwind CSS.
- Implemented dynamic recommendation cards and loading states to improve the user experience.
- Worked with asynchronous API-driven application behavior and frontend state.
- Focused on responsive, functional interfaces while maintaining a polished user experience.

Hospitality Specialist — *Residence Inn / Courtyard Marriott* Jan 2021 – May 2021

- Delivered customer-focused service in a high-volume environment.
- Developed communication, problem-solving, prioritization, and customer-service skills.
- Learned to remain effective while handling multiple requests and changing priorities.

SELECTED PROJECTS

1337 Wing — *Full-Stack Software, Infrastructure & Security Project*

Live Demo: https://1337wing.taila4d3fb.ts.net/

GitHub: https://github.com/agentred1999/1337-Wing

- Designed and developed a full-stack e-commerce-style web application using React and Vite.
- Implemented product cards, product browsing, shopping-cart functionality, interactive interfaces, and a live Hacker News feed.
- Built the backend using Node.js, Express, and PostgreSQL.
- Implemented JWT-based authentication for protected functionality.
- Used bcrypt password hashing to protect user credentials.
- Implemented rate limiting to reduce abuse of application endpoints.
- Used parameterized database queries to reduce SQL injection risk.
- Configured SSH key authentication for secure server administration.
- Configured UFW firewall rules to restrict unnecessary network access.
- Configured Caddy as a reverse proxy for the web application.
- Deployed the application on self-hosted Linux infrastructure.
- Used systemd to manage services and automate application startup.
- Used Tailscale for secure networking and remote access infrastructure.
- Implemented backups and recovery procedures for the self-hosted environment.
- Performed server administration, deployment, troubleshooting, configuration, and security hardening independently.
- Applied accessibility practices aligned with WCAG principles.
- Designed the project around the philosophy of building functional technology first while considering usability and security throughout.

Ultraverse — *Web Development Project*

Portfolio: https://agentred1999.github.io/agentred1999-internship/

GitHub: https://github.com/agentred1999/agentred1999-internship

- Developed a web-based project demonstrating frontend development, responsive design, and interactive user interfaces.
- Applied HTML, CSS, and JavaScript to build functional web experiences.
- Used Git and GitHub for source control and project management.

- Focused on usability, responsive layouts, and practical frontend implementation.

NFT Marketplace — *Frontend / Firebase Project*
GitHub: https://github.com/agentred1999/agentred1999-internship

- Built a React and Firebase marketplace interface.
- Implemented countdown timers, author profiles, and interactive marketplace components.
- Developed shimmer skeleton loading states to provide visual feedback while content loads.
- Focused on responsive browsing experiences and modern interface behavior.

Personal Homelab & Linux Infrastructure — *Independent Systems & Cybersecurity Learning Environment*

- Maintain hands-on Linux and Raspberry Pi infrastructure for experimentation, development, networking, and cybersecurity learning.
- Work with Raspberry Pi hardware and Linux-based systems to explore self-hosting and network services.
- Use Linux systems for server administration, command-line work, networking, troubleshooting, and development.
- Experiment with network security, DNS, SSH, system administration, and infrastructure configuration.
- Use hardware and software troubleshooting skills to diagnose problems across multiple computing environments.
- Explore cybersecurity concepts through controlled personal infrastructure and homelab experimentation.

TECHNICAL PROJECT EXPERIENCE

Linux Server Deployment · Raspberry Pi Administration · Self-Hosted Web Applications · PostgreSQL Database Deployment · Reverse Proxy Configuration · Authentication Systems · Firewall Configuration · SSH Administration · Networking · DNS · TCP/IP · Subnetting · System Services · Backup Systems · Web Application Security · Hardware Troubleshooting

EDUCATION

B.S., Hospitality Administration and Management — *University of Houston* Aug 2021 – Dec 2026

Front-End Development Certificate — *Frontend Simplified* May 2026 – Aug 2026

ADDITIONAL TECHNICAL INTERESTS

Cybersecurity · Linux · Open Source Software · Self-Hosted Infrastructure · Raspberry Pi · Networking · Web Application Security · Computer Hardware · Systems Administration · Homelab Development · AI Applications · Accessibility · User Experience